Interactive Energy Analytics Dashboard: Visualizing India's Renewable Energy Transition (2019–2024) Using Microsoft Power BI

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Abstract—India's ambitious renewable energy transition, targeting 500 GW of non-fossil fuel-based electricity capacity by 2030 under the National Action Plan on Climate Change, demands robust analytical tools capable of transforming large, multi-dimensional energy datasets into interpretable insights for policymakers, grid planners, and researchers. This paper presents the end-to-end design, development, and analysis of a two-page interactive Microsoft Power BI dashboard built on the Ember Climate India Monthly Full Release dataset, comprising 238,200 records spanning January 2019 to December 2024 across all 28 states and 8 union territories of India. The analytical pipeline encompasses data ingestion from CSV format, preprocessing and type conversion in Power Query Editor, and five objective-based analyses: (1) renewable capacity growth tracking, (2) state-wise capacity ranking, (3) fuel mix composition analysis, (4) electricity generation trend analysis, and (5) CO_2 intensity reduction monitoring. The Energy Overview dashboard page integrates four KPI cards displaying installed capacities in MW, two line charts for generation and fuel-wise capacity trends, a stacked area chart with an analytics trend line, and a donut chart for energy mix visualization. The State-wise Analysis page incorporates a geographic bubble map for spatial capacity distribution, a ranked horizontal bar chart, and a state-wise pie chart. Three interactive slicers—for state selection, date range filtering, and energy variable selection—provide dynamic cross-filtering across all visuals simultaneously. Key findings include a threefold increase in solar installed capacity from 2019 to 2024, total renewables reaching 27.13M MW-months, coal retaining a 65.94% share of the generation mix with solar emerging at 15.27%, and a measurable decline in CO_2 intensity confirming India's accelerating clean energy transition.

Keywords—Power BI; renewable energy; interactive dashboard; data visualization; India electricity; solar capacity; wind energy; CO_2 emissions; energy analytics; Ember Climate dataset; business intelligence

I. INTRODUCTION

India stands as the world's third-largest electricity consumer and one of its fastest-growing energy markets, with demand projected to more than double by 2040 relative to 2019 levels according to the International Energy Agency [1]. The nation's dual imperative—meeting rapidly growing energy demand driven by industrial expansion, urbanization, and rising living standards while simultaneously transitioning toward cleaner fuel sources to meet its Nationally Determined Contributions under the Paris Agreement—requires evidence-based decision-making supported by rigorous data analysis and clear visual communication.

Traditional static reports and spreadsheet-based analysis fall short in enabling exploratory, real-time analysis of multi-dimensional energy datasets that span all Indian states, multiple time periods, and diverse energy types. Business Intelligence tools such as Microsoft Power BI offer drag-and-drop visualization, cross-filtering interactivity, built-in analytics including trend forecasting, and no-code dashboard construction, making them accessible to both technical and non-technical stakeholders including government planners and energy regulators [2].

The proliferation of high-resolution monthly electricity datasets made publicly available by organizations such as Ember Climate has created significant opportunities to apply data analytics to real-world energy challenges. The Ember Climate India Monthly Full Release dataset, covering installed capacity, electricity generation, and power sector emissions from 2019 to 2024 across all Indian states, provides an ideal foundation for constructing a comprehensive, state-level energy analytics dashboard.

This paper presents a two-page interactive Power BI dashboard using this dataset. The primary objectives of this study are: (i) to design an interactive, multi-page Power BI dashboard that enables exploratory analysis of India's energy data; (ii) to perform five distinct objective-based analyses covering capacity, generation, fuel mix, and emissions; (iii) to implement dynamic cross-filtering slicers that enable state-level and temporal subsetting; and (iv) to derive policy-relevant insights from the visual analysis that contribute to understanding India's energy transition progress from 2019 to 2024.

II. LITERATURE REVIEW

Suganthi and Samuel conducted a comprehensive review of energy demand forecasting models, emphasizing that visualization of energy trends is a critical prerequisite to forecasting model development, as it reveals structural patterns, seasonal cycles, and anomalies that inform feature engineering decisions [3]. Their findings directly motivate the multi-chart visualization approach adopted in the present dashboard.

The World Bank's Regulatory Indicators for Sustainable Energy (RISE) reports have consistently emphasized that data analytics capacity is a critical enabler of renewable

energy deployment in developing economies, recommending the integration of exploratory visual analysis with policy evaluation frameworks [4]. This recommendation underpins the present study's decision to combine multiple chart types with interactive slicers that support policy-relevant state-level comparisons.

Several recent studies have demonstrated the effectiveness of Power BI and similar BI tools for energy sector analysis. Interactive dashboards have been shown to significantly reduce the time required for energy planners to identify consumption anomalies, capacity gaps, and regional disparities compared to static report-based approaches. The cross-filtering capability of Power BI, which allows a selection in one visual to simultaneously filter all other visuals on the page, is particularly valuable for multi-dimensional energy datasets where state, time period, and fuel type interact in complex ways [2].

India's renewable energy literature has documented the rapid growth of solar photovoltaic capacity driven by the National Solar Mission, declining equipment costs, and state-level auction mechanisms [5]. However, comprehensive state-level visual analysis of the full 2019–2024 period covering capacity, generation, and emissions simultaneously remains limited. The present study addresses this gap by constructing a dashboard that integrates all three dimensions with interactive filtering.

III. DATASET DESCRIPTION

The primary dataset is the Ember Climate India Monthly Full Release in long format (india_monthly_full_release_long_format.csv), containing 238,200 rows and 13 columns. Each record corresponds to a unique combination of state, date, energy category, and variable. The dataset was loaded into Power BI Desktop and found to cover January 2019 to December 2024 across all 28 states and 8 union territories, plus a national aggregate row labelled India Total. Table I summarizes the 13 columns.

TABLE I. Dataset Column Schema

Column Name	Data Type	Description
Country	str	Always India
State	str	State or Union Territory name
State code	str	Two-letter state abbreviation
Date	str → datetime	DD-MM-YYYY monthly timestamp
Category	str	Capacity / Generation / Emissions
Subcategory	str	Fuel / Aggregate fuel / Total
Variable	str	Solar, Wind, Hydro, Coal, etc.
Unit	str	MW / GWh / $ktCO_2$ / gCO_2/kWh
Value	float64	Numeric measurement value
YoY % change	float64	Year-over-year percentage change
YoY absolute change	float64	Year-over-year absolute change

Three primary analytical categories exist within the dataset. The Capacity category records installed generation capacity in megawatts (MW) for each fuel type and state. The Electricity generation category records monthly electricity output in gigawatt-hours (GWh). The Power sector emissions category records CO_2 emissions in kilotonnes ($ktCO_2$) and CO_2 intensity in grams of CO_2 per kilowatt-hour (gCO_2/kWh). The YoY percentage change and YoY absolute change columns provide pre-computed year-over-year growth metrics, enabling direct use as KPI fields without additional DAX calculations.

IV. METHODOLOGY

A. Data Loading and Preprocessing

The CSV dataset was imported into Power BI Desktop via the Home → Get Data → Text/CSV pathway. Power BI automatically detected most column data types; however, the Date column required manual conversion from string format (DD-MM-YYYY) to the Date data type in Power Query Editor to enable correct chronological sorting and date hierarchy creation. The Value column was explicitly set to Decimal Number type to ensure accurate SUM aggregations. The State column was verified to contain consistent naming across all records. No rows required removal as the dataset contained clean, well-structured records throughout all 238,200 entries, with null values present only in YoY change columns for the earliest dates, which is expected behavior for year-over-year computations.

B. Five Objective-Based Analyses

Five distinct analytical objectives were defined prior to dashboard construction, each mapped to one or more specific visuals and filter configurations:

- Objective 1 — Renewable Capacity Growth Over Time: Track the year-wise installed capacity trajectory of Solar, Wind, and Hydro sources from 2019 to 2024. Implemented as a stacked area chart with Date on the X-axis (Year level), Sum of Value on the Y-axis, Variable in the Legend, filtered to Category = Capacity and Variable ∈ {Solar, Wind, Hydro}. An analytics trend line was added via the Analytics pane.
- Objective 2 — State-wise Capacity Ranking: Identify and rank Indian states by total installed renewable capacity. Implemented as a horizontal clustered bar chart with State on the Y-axis and Sum of Value on the X-axis, filtered to Category = Capacity, sorted descending by value.

- Objective 3 — Fuel Mix Composition: Determine the proportional share of each fuel source in total installed capacity. Implemented as a donut chart with Variable in the Legend and Sum of Value as the measure, filtered to Category = Capacity and Subcategory = Fuel to isolate individual fuel types.
- Objective 4 — Electricity Generation Trend: Analyze year-wise national electricity generation patterns. Implemented as a line chart with Date (Year) on the X-axis and Sum of Value (GWh) on the Y-axis, filtered to Category = Electricity generation.
- Objective 5 — CO_2 Intensity Reduction Tracking: Monitor the carbon intensity of India's electricity grid over the study period. Implemented as a line chart filtered to Category = Power sector emissions and Variable = CO2 intensity, with units in gCO_2/kWh.

C. Dashboard Page 1 — Energy Overview

The first dashboard page, titled 'India Renewable Energy Dashboard 2019–2024', provides a national-level summary of India's renewable energy landscape. The page layout positions four KPI Cards across the top row, displaying Solar Capacity (10.17M MW), Hydro Capacity (8.35M MW), Wind Capacity (6.87M MW), and Renewables Capacity (27.13M MW). Each card uses visual-level filters on Category = Capacity and the respective Variable value to isolate the correct data subset.

Below the KPI row, two line charts occupy the second row: the left chart shows the Renewable Capacity Growth trend (MW) over years, while the right chart shows the Electricity Generation Trend (GWh). Both use Date at the Year hierarchy level on the X-axis and Sum of Value on the Y-axis. The third row contains the stacked area chart (Solar, Wind & Hydro Capacity Over Time) with a dotted analytics trend line overlay demonstrating the overall upward trajectory, and the Energy Mix by Fuel Type donut chart revealing the proportional composition of the generation mix. Three slicers—State, Date range slider, and Variable checklist—are positioned on the right side of the page. Fig. 1 presents the complete Energy Overview page.

Fig. 1. Energy Overview Dashboard (Page 1). Four KPI cards, stacked area chart with trend line, generation line chart, and donut chart revealing Coal at 65.94% and Solar at 15.27% of total capacity mix.

D. Dashboard Page 2 — State-wise Analysis

The second dashboard page, titled 'State-wise Renewable Energy Analysis 2019–2024', focuses on geographic and comparative state-level analysis of renewable energy distribution across India. The page is organized into three main visuals alongside a State dropdown slicer for interactive filtering.

The geographic bubble map (Renewable Capacity by State) uses the State field as the Location and Sum of Value as the bubble size metric, filtered to Category = Capacity. The map provides immediate spatial context for the distribution of renewable capacity, with larger bubbles concentrated in western and southern India. The State-wise Capacity Distribution pie chart displays the proportional contribution of each state, clearly showing the dominance of Gujarat, Maharashtra, Tamil Nadu, Rajasthan, Karnataka, and Andhra Pradesh. The Top States — Total Energy Capacity bar chart ranks all states in descending order of Sum of Value, providing a precise numerical comparison to complement the geographic visualization. Fig. 2 shows the complete State-wise Analysis page.

Fig. 2. State-wise Analysis (Page 2). Geographic bubble map, pie chart, and ranked bar chart confirm Gujarat and Maharashtra as top renewable capacity states after the India Total aggregate.

E. Interactive Slicers and Cross-Filtering

A key design objective of the dashboard was to enable dynamic, real-time exploration without requiring separate filtered reports for each state or time period. This was achieved through three synchronized slicers on each page. The State dropdown slicer allows selection of any individual state or union territory, instantly updating all charts to display only that entity's data. The Date Between-style range slider spans January 2019 to December 2024, enabling temporal subsetting to any sub-period within the study range. The Variable checkbox slicer allows filtering by specific fuel types such as Solar, Wind, Coal, or Hydro. All three slicers are linked to every visual on their respective page through Power BI's default cross-filtering mechanism, ensuring synchronized updates across the entire dashboard.

V. RESULTS AND DISCUSSION

The five objective-based analyses collectively reveal a consistent and statistically interpretable picture of India's energy transition from 2019 to 2024. Table II summarizes the key findings from each analytical objective.

TABLE II. Summary of Five Objective-Based Analytical Findings

Analysis Objective	Key Finding
1. Renewable Capacity Growth	Solar capacity increased ≈3× (2019–2024)

2. State-wise Ranking	Gujarat, Maharashtra, Tamil Nadu lead
3. Fuel Mix Composition	Coal 65.94%, Solar 15.27%, Hydro 12.53%
4. Generation Trend	Consistent upward growth every year
5. CO_2 Intensity	Declining trend confirms cleaner grid

Solar installed capacity demonstrated the most rapid growth of all renewable sources, increasing approximately threefold from 2019 to 2024. The stacked area chart in Fig. 1 clearly illustrates Solar's expanding dominance over Wind and Hydro in annual capacity additions, consistent with India's National Solar Mission targets and the dramatic decline in photovoltaic equipment costs over the study period [5]. The KPI cards confirm total Solar at 10.17M MW-months, Wind at 6.87M, Hydro at 8.35M, and aggregate Renewables at 27.13M, demonstrating the cumulative scale of India's renewable capacity build-out across all states and years.

The donut chart in Fig. 1 reveals that Coal retains a dominant 65.94% share of the installed capacity mix, while Solar at 15.27% has emerged as the second-largest component—a transformation from 2019 when Solar's share was considerably smaller. Hydro contributes 12.53% and Wind 6.87%, indicating that both sources have grown in absolute terms but at a slower pace than Solar. The continued dominance of Coal underscores the scale of the transition still required to meet India's 2030 targets.

The State-wise Analysis page (Fig. 2) identifies Gujarat, Maharashtra, Tamil Nadu, Rajasthan, Karnataka, and Andhra Pradesh as the six leading states in renewable capacity. The geographic bubble map makes this spatial concentration immediately interpretable, showing a pronounced clustering of large bubbles in western and southern India. This geographic pattern reflects the superior solar irradiance in Rajasthan and Gujarat, and strong wind resources along the western and southern coastlines [4].

The CO_2 intensity line chart shows a consistent declining trend across the 2019–2024 period, confirming that the substantial renewable capacity additions are translating into measurably lower carbon intensity of the electricity grid. This is a critical policy metric, as it demonstrates that India's renewable investment is delivering tangible decarbonization outcomes rather than simply adding to total capacity alongside unchanged fossil generation.

The electricity generation trend chart shows consistent upward growth across the study period, with every year from 2021 onward exceeding the prior year level. The interactive cross-filtering capability confirmed that selecting individual states dynamically updates all visuals simultaneously, enabling granular state-level policy evaluation. For example, selecting Rajasthan in the State slicer immediately reconfigures all charts to display Rajasthan's solar-dominated capacity profile, illustrating how the dashboard eliminates the need for separate per-state static reports.

VI. CONCLUSION

This paper presented the end-to-end design and development of a two-page interactive Microsoft Power BI dashboard for analyzing India's renewable energy transition using 238,200 records of monthly energy data spanning January 2019 to December 2024. Five objective-based analyses covering renewable capacity growth, state-wise capacity ranking, fuel mix composition, electricity generation trends, and CO_2 intensity reduction were systematically performed and visually presented through six distinct chart types and three dynamic interactive slicers.

Key findings from the dashboard analysis include: (i) solar installed capacity increased approximately threefold from 2019 to 2024, demonstrating Solar's emergence as India's fastest-growing energy source; (ii) total aggregate renewable capacity reached 27.13M MW-months across the study period; (iii) Coal retains a 65.94% share of total installed capacity while Solar has risen to 15.27%, signaling a meaningful but incomplete transition; (iv) Gujarat, Maharashtra, Tamil Nadu, Rajasthan, Karnataka, and Andhra Pradesh account for the majority of India's renewable capacity, reflecting geographic concentration driven by solar and wind resource endowments; and (v) a consistent decline in CO_2 intensity across the study period confirms that renewable capacity additions are delivering measurable grid decarbonization outcomes.

The interactive dashboard with three synchronized cross-filtering slicers—State, Date range, and Variable—provides a reusable analytical template that enables state-level and temporal exploration without the need for separate static reports. This design approach is directly applicable to energy regulators, grid planners, and policy researchers requiring rapid, flexible access to multi-dimensional energy data.

Future work could incorporate DAX measures for custom KPIs such as renewable penetration percentage and year-over-year capacity addition rates. Integration with Power BI Service and scheduled dataset refresh would support real-time monitoring as new monthly data are released by India's Central Electricity Authority. Extension to include electricity demand data, tariff information, and weather covariates would enable a more comprehensive analysis of the drivers of India's energy transition.

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